

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated
for lumbosacral transitional anatomy**

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Purpose: The gold standard for naming a vertebra at the lumbosacral junction is counting caudally from C2 on whole-spine imaging. In practice, a lumbar surgical case is planned on lumbar-only imaging (T12 to S1) without C2, making the gold standard method impossible. Abdominopelvic computed tomography (CT) holds the lumbar-only imaging span plus the lowest ribs and pelvis, making it a reasonable substitute for the preoperative view. When a lumbosacral transitional vertebra (LSTV) alters the count, however, the local anatomy is much more ambiguous. A person who has four rib-free lumbar vertebrae may have an L1 with a lumbar rib or an L5 assimilated to the sacrum, and a person who has six rib-free lumbar vertebrae may have an L6, a T12 with aplastic ribs, or a lumbarized S1. CTSpinoPelvic1K asks whether local morphology can resolve that ambiguity without counting caudally from C2. Two public label sets already covered this imaging, CTSpine1K’s vertebrae and CTPelvic1K’s pelvis, on the same colonography patients. This release joins them in one dataset with 802 CT records, per-level ribs and femora, and lumbar levels anchored on the lowest rib-bearing vertebra and the sacrum. It also introduces classes for L6, T13 and lumbar ribs, so anomalies are accurately recorded and not forced into whichever normal count they happen to resemble.

Acquisition and Validation Methods: Records pair CTSpine1K and CTPelvic1K labels on each patient’s bone-richest acquisition under a VerSe-native scheme. Validation covered geometric invariants (802/802 pass), rib–vertebra incidence across 5,749 ribs (0.035% offset), and spinopelvic measures matching published values.

Data Format and Usage Notes: NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a loader; archived at <https://doi.org/10.5281/zenodo.22139642>.

Potential Applications: Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar cohort, surgical planning research (377 records prone). *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus on 33 records.

6 I. INTRODUCTION

7 The spine typically has seven cervical, twelve thoracic, and five lumbar vertebrae. A
8 vertebra is classified by where it falls in that sequence. Transitional variants sit at the two
9 ends of the lumbar column, with the thoracolumbar transitional vertebra (TLTV) above
10 and the lumbosacral transitional vertebra (LSTV) below. Both transitional variants disturb
11 identification by changing what the border vertebra and its ribs *look like*. For example, a
12 thoracolumbar border vertebra may carry a full rib, a hypoplastic stump, or none at all, and
13 a lumbosacral one may be partly or completely assimilated to the sacrum (*sacralization*),
14 with an anteriorly wedged body, a reduced disc beneath it and hypoplastic facet joints, or
15 separated from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height
16 disc¹. In addition, they change how many vertebrae each region holds: eleven or thirteen
17 thoracic and four or six lumbar.

18 Together, these transitional variants make it difficult to classify the vertebra at these
19 borders definitively. Each of those readings yields the same count (Fig. 2), and the morphol-
20 ogy that would separate them is not always decisive to human readers. Whether it differs
21 enough for a trained classifier to separate them, or at least to assign each reading a probable
22 classification, is the question this release aims to answer. The gold standard for numeration
23 in transitional anatomy is whole-spine imaging, counting caudally from C2², which deter-
24 mines how many vertebrae and how many rib pairs the column holds. No spine-limited
25 acquisition can supply either count. LSTV is common, reported at 10–29% in the general
26 population³ and in 16.3% of a whole-spine CT series⁴, and wrong-level surgery is its most
27 serious consequence.

28 Existing public collections cannot address this, and not for lack of size (Table I). Prior
29 spine datasets omit the pelvis and pelvic datasets do not number the vertebrae. TotalSeg-
30 mentator⁵ has no class for a sixth lumbar vertebra or for a rib on a lumbar vertebra, and
31 VerSe⁶, the one collection with an L6 class, stops at the sacrum and carries no ribs. As
32 such, none of these collections can record a spine with six lumbar vertebrae together with
33 both of its borders, and a segmenter trained on them must shift the whole column by a
34 level or absorb a vertebra into its neighbor when it encounters an enumeration anomaly.
35 This problem has now been measured on CT, where prior labeling methods assigned every
36 vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS⁷ raised

that to 99%⁸. Its weights label L6 and T13 on CT, but its 1,536-scan CT cohort is in-house and unreleased, so what exists is a capability, not a dataset.

When considering the intraoperative side, LevelCheck registers the intraoperative radio-graph to the preoperative CT and projects the CT’s vertebral labels onto it^{9–11}. Those labels, however, are placed by hand and verified by the surgeon⁹, so the method assumes a correctly labeled CT and does not supply one itself. As such, a dataset carrying the anomaly classes is vital for automating that step.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and classifies each structure an enumeration anomaly produces, including a sixth lumbar vertebra and lumbar ribs. A scheme without those classes is unable to adequately describe anatomic anomalies.

The project began with the question whether a vertebra can be named from the anatomy on the images a lumbar surgical case is actually planned on. By guideline those images are lumbar-only^{12–15} and whole-spine coverage is reserved for trauma with an identified injury^{16,17} and for deformity radiographs. An abdominopelvic CT runs from the diaphragm to the pelvic floor, so it captures that span with the lowest ribs and the whole pelvis, while lacking C2. Every record comes from one prospective trial protocol, ACRIN 6664^{18,19}, which scanned 802 patients aged 50 and over, supine and prone, on multidetector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge, it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol. The comparators, by contrast, pool collections, are multi-site by design, or are routine clinical scans under many protocols (Table I). With the protocol for this study fixed, scanner effects on a model become measurable rather than confounded. The downside is the restriction to an abdominal field of view, in which only the lowest thoracic levels are present (Sec. V).

The gap was evident. CTPelvic1K²⁰ and CTSpine1K²¹ each annotated the COLONOG collection under radiologist supervision for the same patients, with one annotating the pelvis and the other annotating the vertebrae. The two had remained separate, although joining them needs no new imaging and no new radiologist. It was not a file merge, because each annotation had to be traced back to the series it was drawn on (Sec. II.A). Once completed, that merge yielded a combined spine-and-pelvis frame for 802 patients, and the bones neither set had, ribs, femora and hardware, could be annotated against it rather than from scratch.

69 I.A. Vertebra labeling when the scan cannot settle the count

70 The main limitation is that no scan in this dataset contains C2, so the gold standard
71 conventional count cannot be performed. This is a property of abdominal imaging rather
72 than of the annotation itself. A thirteenth thoracic vertebra and an L1 with a lumbar rib are
73 different phenotypes, distinguished as separate subtypes in cadaveric classifications of the
74 thoracolumbar junction^{22,23}. The first is an additional rib-bearing vertebra positioned above
75 five lumbar vertebrae, so the column has 25 presacral vertebrae. The second is a rudimentary
76 rib on a lumbar-type L1 in a column with the standard 24 presacral vertebrae. Neither is
77 separable by counting from the last rib-bearing vertebra or from the sacrum because four
78 rib-free vertebrae can fit a lumbar rib, a cranially shifted T13 or an L5 assimilated to the
79 sacrum, and six rib-free vertebrae can fit an aplastic twelfth rib or a sixth lumbar vertebra.
80 A *stump rib* is a hypoplastic rib that is a key indicator of a thoracolumbar transitional
81 vertebra. On whole-spine CT scans, rib anomalies travel with the lumbosacral border and
82 hypoplastic twelfth ribs occur with sacralization and lumbar ribs with lumbarization.⁴

83 In this release, every vertebra carries the identifier its radiologist-sourced annotation
84 assigned as the ground truth, corrected where the source was wrong. In addition, every
85 released measurement is expressed against the two landmarks present in essentially every
86 abdominal CT scan: the sacrum inferiorly and the lowest rib-bearing vertebra superiorly.
87 This makes the measurements positional rather than nominal, with a vertebra being classified
88 by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as
89 a fraction of the rib above it, and a transverse process by its span and its distance from
90 the ala. None of these values shift with the name a reader gives the junction, meaning that
91 cases that disagree about the name remain comparable.

92 *But the vertebrae themselves are known to differ.* Level-specific morphometry is long es-
93 tablished in the literature: pedicle width and height change systematically from the thoracic
94 to the lumbar spine,²⁴ vertebral body, endplate and canal dimensions differ by level,^{25–27} and
95 the transverse process and the iliolumbar ligament mark the last lumbar vertebra indepen-
96 dently of any count.^{1,28} If those differences hold at the boundary, the phenotype is decidable
97 locally, and on this dataset it is. Separating T12 from L1 on shape alone gives an area under
98 the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. This was calculated using a logis-
99 tic regression on five released per-level measures and six ratios between them. Each measure

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it³⁰.

Collection	Scans	Count	L6	Sacrum	S1	Pelvis	Ribs	L. rib	Femora	Grade
CTSpine1K ²¹	1,005									
CTPelvic1K ²⁰	1,184									
VerSe ⁶	374									
RibSeg v2 ³¹	660									
TotalSegmentator ⁵	1,204									
CTSpinoPelvic1K	802									

was divided by the patient’s median across their levels, so that size, which separates thoracic from lumbar trivially and says nothing at the junction, is removed. Folds are grouped by case and the curve is taken on pooled out-of-fold scores (`test_morphometric_separability.py` in the released code). Schinz et al. reach the same conclusion from the other side: on 1,242 whole-thoracolumbar CTs a shape-based labeling of the junction matched nerve morphology in every case, against 92.6–97.2% for counting- and rib-based rules.²⁹

II. ACQUISITION AND VALIDATION METHODS

II.A. Sources, and why a crosswalk was necessary

CTSpine1K’s 1,005 volumes and CTPelvic1K’s 1,184 come from four and seven collections of CT obtained for different indications. Only the Cancer Imaging Archive (TCIA) CT colonography collection (COLONOG)^{18,19} is common to both, and it is the largest in each: a subset of a prospective screening trial of asymptomatic patients aged fifty and over, acquired under one protocol. Every COLONOG patient was scanned prone and supine at a minimum, with 12 exceptions among the 825 patients the sources drew on. CTSpine1K states that one position was chosen at random by a released script; that script, since deleted but preserved in the repository’s history, writes every series holding twelve or more files to the same filename, so the volume retained is whichever series the directory walk reached last. No code was released with CTPelvic1K. Consistent with two independent choices, the sources annotated the same series for 342 of the 693 patients both annotated (49.4%), and the released records mix positions and reconstructions accordingly: 422 supine, 377 prone, three

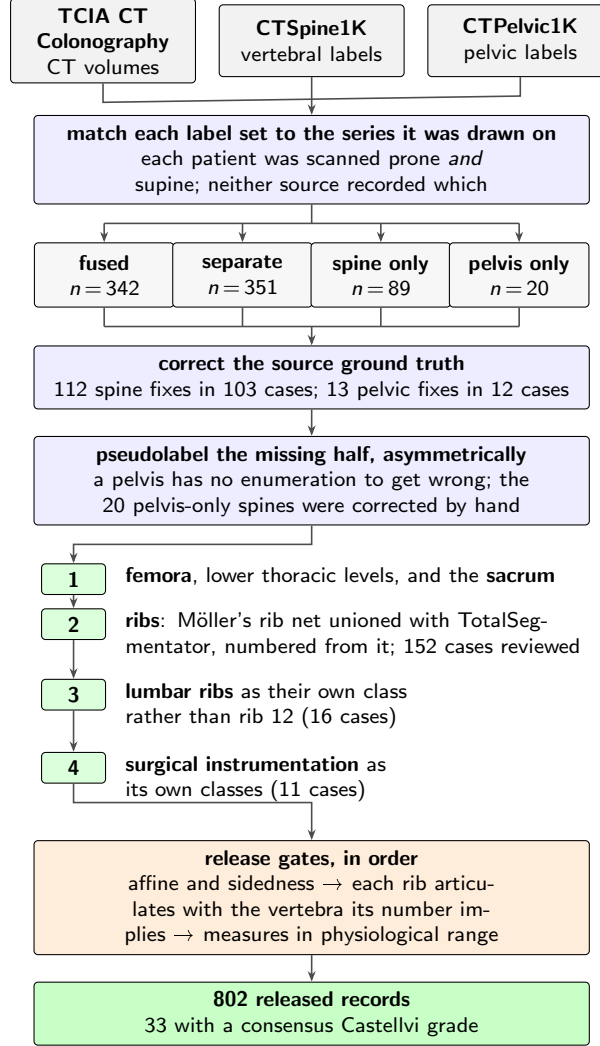


FIG. 1 How the dataset was built. n counts scans, or corrections, at each step.

120 decubitus; 562 at 1.0 mm, 232 at 1.25 mm and eight at other slice thicknesses.

121 Recovering the archived series each annotation was drawn on is nontrivial because the DI-
 122 COM attributes that identify an acquisition, the patient, study and series instance identifiers
 123 and the patient position,³² have no counterpart in the NIfTI-1 header³³ and are discarded on
 124 conversion.³⁴ CTSpine1K's `Path.csv` names the TCIA directory each label was converted
 125 from, but it carries no series identifier, names no position for 217 of the 782 annotated
 126 records, and for seven indicates an acquisition other than the one whose bone the label over-
 127 laps best. CTPelvic1K released no imaging and encoded the mapping in its mask filenames,
 128 `dataset2_PatientID_SeriesNumber_undocumented_[tag_]mask_4label.nii.gz`. One of
 129 the 714 masks names a patient with no archived series; of the other 713, the first two fields re-
 130 solve 702 to the series selected here by bone overlap, five to a `SeriesNumber` two series share,
 131 and six to one no series carries. The third field matches nothing, and the fourth, on 32 masks,

132 mixes phenotype (`sacralization` 7, `semisacralization` 2, `hard_sacralization` 1), diffi-
133 culty (`hard` 8, `veryhard` 2), image condition (`lowdose` 6; `metal`, `crop`, `IntestinalCalculus`
134 once each) and two unexplained tags (`dqjoint` 2, `ydjoint` 1) in one string.

135 The crosswalk (Fig. 1) therefore resolved every annotation, not only the 27.7% with
136 no stated position, by thresholding each of the patient’s volumes at bone attenuation and
137 keeping the series the mask overlaps most; where a source did name a series, 558 of the
138 565 CTSpine1K records naming a position and 702 of the 713 CTPelvic1K masks naming
139 an archived patient agree with it. The result is 342 records whose spine and pelvic labels
140 land on the same series (*fused*), 351 whose labels land on different series of one patient
141 (*separate*), 89 with a spine label only and 20 with a pelvic label only. Turning a patient
142 between prone and supine alters lumbar alignment and the seating of the pelvis against it³⁵,
143 so a mask drawn on one series does not transfer rigidly to the other, and while a patient’s
144 two annotations lie on different scans no spinopelvic measurement can be taken across them.

145 II.B. Densification, and why it is asymmetric

146 A rigid carry-over of each separate patient’s own pelvic annotation onto the spine’s series
147 is implemented in the released code but was not adopted. Instead, the missing region of
148 every partially annotated record was pseudolabeled by a segmentation model³⁶ trained on
149 those records themselves, the unannotated region marked by nnU-Net’s ignore label so that
150 it contributes nothing to the loss, and predicted out of fold, so no record was completed by
151 a model that had trained on it. The pelvis is thus radiologist-derived on the 362 fused and
152 pelvis-only records and pseudolabeled on the other 440 (Fig. 1). The direction is asymmetric
153 because a pelvis carries no enumeration to get wrong, whereas a pseudolabeled spine must
154 commit to a count and, on a transitional vertebra, commits to the commoner reading; the 20
155 pelvis-only spines were corrected by hand. The pelvic pseudolabels were not scored against
156 withheld ground truth; they pass the release gates every structure passes (Sec. II.F), and
157 the spinopelvic measures that depend on them agree with published supine values.

II.C. Ribs

No public rib annotation covers this cohort, and the two available automatic sources^{5,37} fail in complementary ways.

Möller’s binary rib network³⁷ reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. Here, though, the segmentation itself failed on ribs that were *partially outside* the field of view. A rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. In a colonography cohort the lower ribs are routinely clipped, and those happen to be the ribs the count depends on.

TotalSegmentator⁵, by contrast, does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone, where the most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib class labels, which matches each rib to the vertebra its number implies) has nothing to test.

Given these complementary gaps, the two approaches were combined (Fig. 1), TotalSegmentator supplying the numbering and the medial-clipped cases, Möller the detection where its output is more complete.

The result is a pseudolabel whose review was triaged rather than exhaustive. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with, since a single bone carrying two numbers is a numbering error whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and four were not and are identified in the release. Reviewers were medical students working through a student research consortium³⁸, trained against a written labeling protocol, with every correction attributed to its annotator. Records passing the rule were not individually inspected; the residual rib–vertebra offsets reported in Sec. II.F are the independent check on what the rule missed.

187 II.D. Label scheme and counting anchors

188 The scheme is VerSe-native. Vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12
189 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure
190 takes a fixed identifier above that range, hips (30–31), femora (32–33), thirteen ribs per side
191 (34–46 left, 47–59 right), lumbar ribs (60–61) and surgical hardware (62–68). Identifier 29
192 is retired and unused, for the reason given below. Rib 13 is the true rib of a T13 and is
193 empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are recorded as the
194 distinct phenotypes they are.

195 Two anchors give every record the same positional frame, and Fig. 2 shows both, the
196 lowest rib-bearing vertebra above and the **sacrum** below. The sacrum’s superior end-plate
197 is the landmark sacral slope and pelvic incidence are measured from, and with both brackets
198 present L5 and L6 are distinguishable where a spine-limited view alone would not be.

199 *A separate S1 label was withdrawn, and why.* Earlier releases carved the cranial portion of
200 the sacrum as its own S1 class, on an automatic estimate of the S1–S2 boundary. Measured
201 against the anatomy on all 802 records, that estimate is unreliable: S1’s craniocaudal extent
202 is a median 57.6 mm where a first sacral segment is near 30, only 26% of records fall in a
203 defensible range, and in 222 the plane takes more than half the sacrum. The failures are not
204 merely low cuts. Where the fit degenerates the plane comes out nearly parallel to the sacral
205 long axis and shaves the ventral cortex, so the retained sacrum is left without its anterior
206 wall — wrong for every use of the label, not only for the end-plate. Rendering one such
207 record is what showed this; no summary statistic did.

208 The consequence was confined to the parameters fitted to that surface. Splitting the
209 cohort on whether the carve was anatomically sound separates them cleanly: the 580 sound
210 records give a pelvic incidence of 52.7° and a pelvic tilt of 19.6°, against Hasegawa’s supine-
211 CT 53.4 and 19.2, while the 221 degenerate ones give 68.5 and 33.6, and the end-plate
212 gate described in Sec. II.E fires on 96% of the second group and 4% of the first. The
213 measurement was never at fault; the surface it was given was. So S1 is merged back into
214 the sacrum, restoring the bone the source annotation actually delineated, and the end-plate
215 is fitted to the whole sacrum instead.

216 VerSe⁶ carries L6, whose identifier this release keeps, so its L6 is VerSe’s.

217 **A rib on a lumbar body receives its own class**, and this is the one class here with no

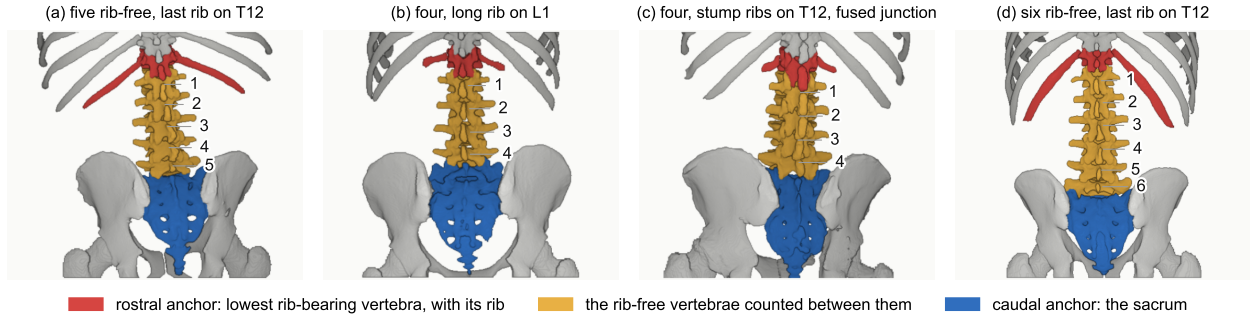


FIG. 2 The two anchors, from the released labels, aligned on the sacrum. Red, lowest rib-bearing vertebra and its rib; blue, the sacrum; yellow, the rib-free vertebrae between. Cases 0704, 0428, 0094 and 0005, chosen by rule as the most ordinary member of each phenotype (`scripts/select_anchor_cases.py`). (b) and (c) both count four. In (b) a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar body is fused to the sacrum (Castellvi IIIb).

published counterpart. A lumbar rib and a stump rib are the same object under two counts, so a scheme that numbers every rib 1–12 leaves the annotator with two bad options: either calling it rib 12, which asserts that the vertebra beneath it is thoracic, the very question at issue, or discarding it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe’s T13 is a vertebra rather than a rib. This scheme records both: VerSe’s T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for a lumbar-type body bearing a rudimentary rib. No record here carries a T13; the sixteen lumbar-rib cases were read as lumbar-type bodies, and where a border vertebra could not be settled by its readers the label stands as read.

Figure 2 shows why the interval is the primitive and the label is derived. Among the records with four rib-free vertebrae, some reach that count from above, through a rib on L1, and some from below, through a lumbosacral segment fused to the sacrum, in Fig. 2(c) together with stump twelfth ribs; the count alone cannot tell them apart, and the release can because ribs, rib lengths and the lumbosacral junction are all labeled.

Sixteen records carry a lumbar rib, thirteen bilateral and three unilateral, all on the right, too few to speak to side preference.

Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the eighteenth was not graded. The source transitional label (pelvic where CTPelvic1K flagged one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and

237 semi-sacralization in one, and is unremarkable in the ungraded record. A spine can carry six
238 lumbar-type vertebrae and still be read as a sacralization, depending on where the reader
239 started the count.

240 Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`,
241 `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by
242 counting identifiers 25 and 60–61.

243 II.E. Computational tools

244 *Access.* Every step is performed by open-source code archived with the dataset, together
245 with the reference loader, the label scheme and the quality-control scripts that generated
246 the tables in this manuscript.

247 *Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-
248 level rib numbering used TotalSegmentator⁵ (`total` task, release 2.x) on a single graphics
249 processing unit. The binary rib network is Möller’s³⁷ released weights, applied through the
250 nnU-Net v2 framework³⁶. The pelvic completion for records whose pelvis was absent used
251 a five-fold nnU-Net v2 ensemble applied out-of-fold, so that no record is completed by a
252 model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset
253 descriptor, are released alongside the data (the repository URL identifies the authors; it is
254 on the title page and will be restored here for the camera-ready version). Image handling
255 throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is
256 written, so every label shares its image’s grid and affine exactly.

257 *Measurement.* Dimensions and spinopelvic parameters come from code released with
258 the dataset. A sacral plate whose fitted normal lies more than 60° off the cranial axis is
259 rejected and the case reported missing, since such a fit has found the near-vertical anterior
260 face of the promontory, not the endplate. Gating on the geometry rather than on the value
261 leaves unusual patients in. an open-source measurement toolkit is released separately (the
262 URL identifies the authors and is on the title page) as a reusable implementation measuring
263 in each vertebra’s own frame; no open-source implementation computing these parameters
264 from CT segmentations was found elsewhere.

265 *Generative artificial intelligence.* A large language model (Claude, Anthropic) assisted
266 with writing the analysis and figure code and manuscript refinement after initial drafting. It

was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes. All content was independently reviewed, verified, and interpreted by the authors.

II.F. Validation

Validation has three layers. Geometry is checked first, because a measurement taken on a corpus with a geometric fault still returns a plausible-looking number.

Geometric invariants. Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed.

The check compares *each sided pair separately*. Testing the ribs alone passed all 802 records, four of which carried a `left_hip` label on the patient’s right side; pooling every sided structure into one test still missed one of those four records, because a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

Rib–vertebra incidence. Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined. Both offsets are field-of-view truncations, one at +1 level and one at −1.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 60 or 61 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

Agreement with published values. Pelvic incidence is the strongest of these checks because it is a morphological property rather than a posture; it agrees to within half a degree with the automated supine CT series of Veilleux *et al.*³⁹ and sits below the standing figure, the

TABLE II Spinopelvic measures, mean $\pm$ SD over the 756 of 802 records passing the geometric gates, against two supine CT series that report means.^{35,39} Standing values are 55.0, 41.0, 13.0 and 60.0°.⁴² The end-plate is fitted to the whole sacrum; the gate rejects 14 records where the previous carved-S1 fit lost 239. Pelvic incidence and tilt are right-skewed (0.25 and 0.33), so their medians—51.5 and 15.0°—sit below these means and the modal peak in Fig. 4 sits below both.

Measure	This dataset (supine, $n = 756$)	Veilleux ($n = 200$)	Hasegawa ($n = 24$)
Pelvic incidence	$52.1 \pm 11.1^\circ$	52.1	53.4
Sacral slope	$36.6 \pm 10.6^\circ$	36.5	34.1
Pelvic tilt	$15.5 \pm 7.9^\circ$	15.6	19.2
Lumbar lordosis	$51.7 \pm 13.3^\circ$	—	—
PI – LL mismatch	$0.3 \pm 10.3^\circ$	—	—

direction reported for subjects imaged both ways.⁴⁰ Sacral slope and pelvic tilt are postural, and read three degrees below and four above Veilleux’s asymptomatic subjects; a second supine CT series reports 53.4, 34.1 and 19.2°, within a degree of this cohort on all three.³⁵ Recumbency runs the wrong way to explain the gap, raising slope rather than lowering it, so the residual tracks cohort and not construction.

II.G. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade⁴¹ in the released manifest, 33 cases typed I–IV with the *a/b* unilateral–bilateral qualifier. The grade and the count are *different axes*. Grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the readers disagreed on, type II (articulation) against type III (bony fusion)¹, is the one that matters clinically, and four of the six disagreements were III against II.

II.H. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because an iatrogenic fusion is indistinguishable from a congenital one to a distance

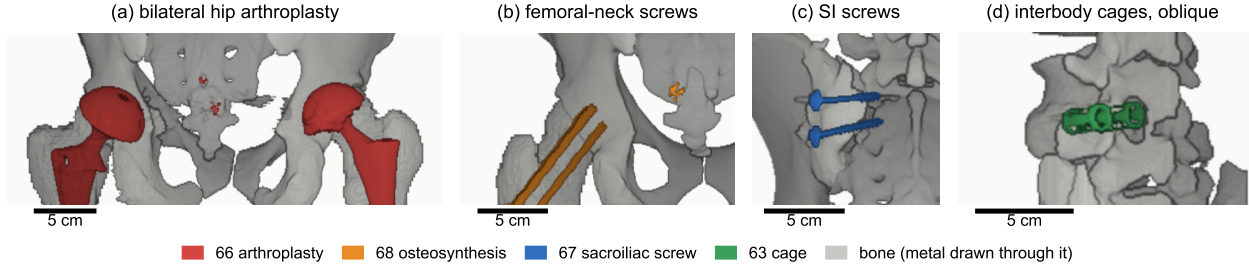


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67, $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

314 **measurement.** A cage-bridged interspace reads as “no gap” exactly as a fused transitional
315 vertebra does.

316 *II.H.0.a. Detection over-calls by a factor of five.* A 2500 HU threshold inside a shell
317 around the labeled skeleton flags 52 of the 802 records. All 52 were reviewed manually,
318 confirming 11 and rejecting 41 (79%) as contrast, calcification and reconstruction artefact.
319 Attenuation does not separate the two groups (both saturate); site and volume do, every
320 confirmed implant at $2,586 \text{ mm}^3$ or above and every rejection at $1,768 \text{ mm}^3$ or below.

321 *II.H.0.b. The subtype block had to be extended.* Identifiers 62–65 name spinal instru-
322 mentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds,
323 so **66 arthroplasty**, **67 sacroiliac screw** and **68 osteosynthesis** were added. Osteosyn-
324 thesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule
325 calls a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis,
326 one sacroiliac fixation and one pair of interbody cages (Fig. 3).

327 *II.H.0.c. Metal outranks bone, and that is a subtraction.* In every confirmed case the
328 implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than
329 adding them. In eight of these cases the femoral head that pelvic incidence and tilt are
330 measured from is an implant.

331 III. DATA FORMAT AND USAGE NOTES

332 Each record is a gzipped NIfTI image/label pair sharing an identical 4×4 affine, canon-
333 icalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM, which is
334 what volumetric segmentation tooling expects.

335 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the

TABLE III Data types and metadata, and the records populating each, from the released manifest.

Field	Records Available	
Identifiers and paths, configuration, match type, image/label alignment check	802	100.0%
Patient position, tube potential, section thickness, kernel	802	100.0%
Transitional label and class, annotation origin, instrumentation flags	802	100.0%
Spine series identifier; bone-fraction check	782	97.5%
Scanner manufacturer and model	768	95.8%
Sex	749	93.4%
Age	709	88.4%
Pelvic series identifier	713	88.9%
Castellvi grade (consensus of two readers)	33	4.1%

data as `splits_5fold.json`. The five validation folds partition all 802 records, so the release provides cross-validation and *not* a held-out test set; users reporting a single held-out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag. Across all 802 records the source transitional label reads lumbarization in 14, sacralization in 17 and semi-sacralization in two; separately, 18 records carry a sixth lumbar-type vertebra, nine carry only four lumbar labels, and 33 carry a consensus Castellvi grade. A record called sacralization by its four lumbar labels enters the same stratum as one called sacralization by the source label, so the two rare strata hold 15 records each and every validation fold carries three of each. That is the most stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records: 393 recorded female and 345 male, 11 carrying DICOM’s *other* value and 53 no sex field, with age present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven outside the female/male

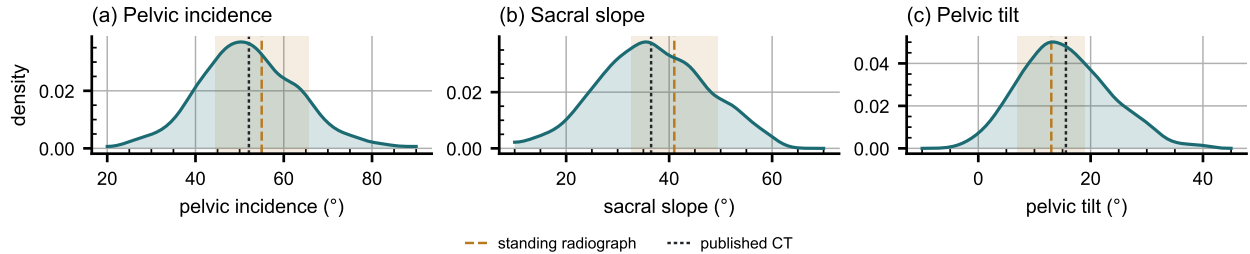


FIG. 4 Spinopelvic measures against standing reference bands (mean $\pm$ SD)⁴².

dichotomy are excluded from sex-stratified measures.

Count-free measures. Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, and its lower mode, below 0.33 of the rib above, marks a stump twelfth rib in 98 of the 789 records with a measurable pair (12.4%, against 12.6% on whole-spine CT⁴); 16 records carry a lumbar rib (2.0%, against 1.5%). The co-occurrence reported there is testable here only where the junction was read, and only within those 789: they hold 14 of the 17 source-labeled sacralizations, and stump ribs accompany four of those 14 and 94 of the other 775 (odds ratio 2.9, $p = 0.08$, Fisher’s exact test). No lumbarization carries a lumbar rib.

Spinopelvic measures (Fig. 4). Pelvic incidence sits on the standing reference, as a morphological property should; slope and tilt sit below and above the published CT values, which Table II places against two series.

Opportunistic measures. Every scan carries a vertebral bone density measurement at no additional dose.⁴³ L1 trabecular attenuation is reported at the published standard site.

V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, making sacral morphology measurable against the lumbar column rather than in isolation, among them the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it, since no trajectory can be planned across a junction whose segments cannot be named.

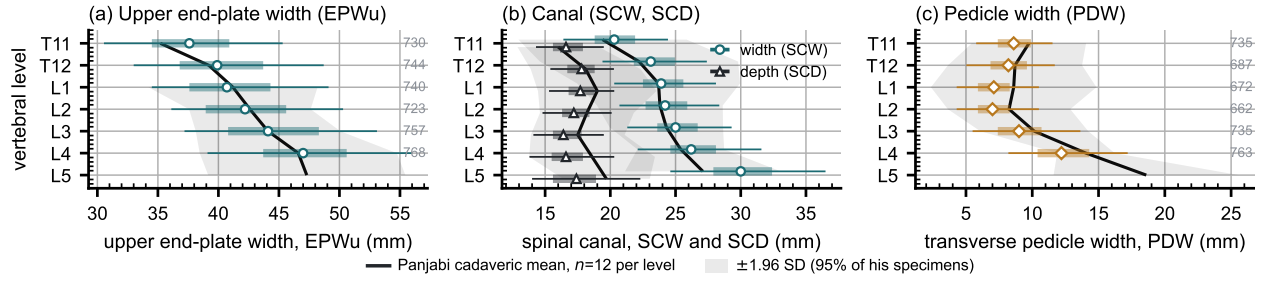


FIG. 5 Morphometry by level, T11–L5: median, interquartile range, 5th–95th percentile, n at right. (a) End-plate width EPWu. (b) Canal width SCW and depth SCD. (c) Pedicle width PDW, both sides averaged. L5 is withheld in (a) and (c); see text. Reference is Panjabi’s cadaveric series^{25,26}, keyed in (a), its SD recovered from the reported SEM; T11–T12 EPWu is digitised from his Figure 4A.

378 *A proxy for the preoperative view.* No public preoperative lumbar CT cohort exists that
 379 we know of, so this release is the closest available stand-in for the view a lumbar case is
 380 planned on: it holds the T12-to-S1 span with the lowest ribs and the pelvis, 377 of its
 381 802 records were acquired prone, the position of posterior lumbar surgery, and it carries
 382 classes for the anomalies that make level identification fail. It is a screening population
 383 at colonography dose, not a surgical series, so it supports method development for level
 384 identification and instrumentation geometry rather than validation of a planning decision.

385 V.A. Three uses the released measurements support

386 *Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one in
 387 3,110 spinal procedures, and among the predominant contributing factors is the transitional
 388 anatomy this cohort was assembled around.^{44,45} Per-level body height, canal width, endplate
 389 width, transverse-process span and wedge ratio are released across all 802 records, and
 390 because the labels were assigned against the twelfth rib rather than by enumeration, a
 391 model trained on them is not learning the convention that fails.

392 *Reference morphometry with its spread.* The values a surgeon consults for endplate and
 393 canal dimensions and pedicle width come from cadaveric series of ten to thirty specimens^{25–27}
 394 redrawn in the textbooks as one curve per level⁴⁶, and a standard error describes the mean,
 395 not the spread of individuals. The same measures are given here from 802 records with
 396 the distribution attached (Fig. 5), reproducing the classical pedicle widening into the lower

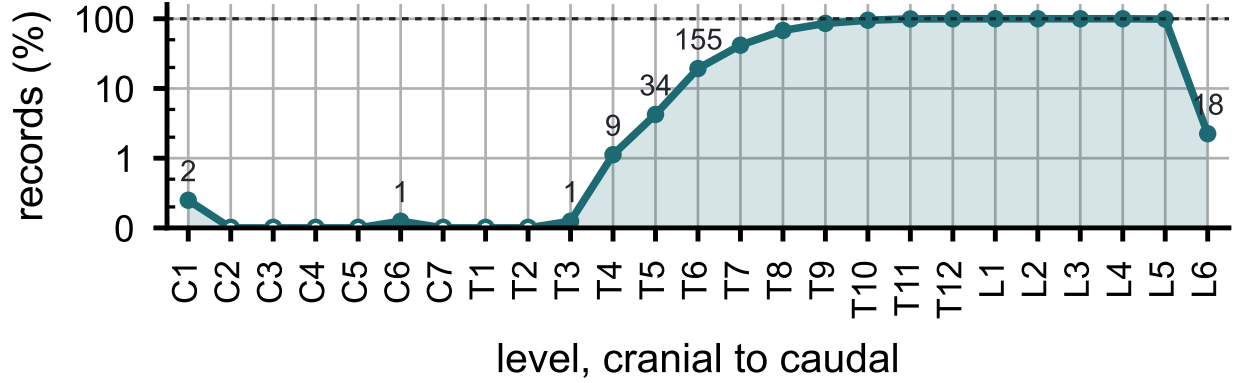


FIG. 6 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

lumbar spine. Both layers of that figure are population intervals rather than confidence intervals, and that distinction is what makes their widths comparable: a confidence interval bounds a parameter, a population interval bounds individuals,⁴⁷ and at $n = 12$ the two differ by $\sqrt{12}$.

Patient-specific models. Patient-specific finite-element modeling for alignment planning and implant design⁴⁸ needs spine, pelvis and femoral heads in one frame, which is this release’s construction, and 351 patients give a within-patient postural change to check a prediction against.

Limitations: coverage and cohort. Thoracic ground truth is field-of-view limited (Fig. 6) and does not reach T1. Postural angles are supine; pelvic incidence is not postural, but it is modality-sensitive, reading below radiography for the same subjects.⁴⁰ End-plate and pedicle width are withheld at L5, where the transverse processes arise in front of the canal wall the body is cut at. Those two and the spinopelvic angles, which rest on the sacral end-plate, are the provisional part of this release; the planned revision takes body and sacral plate from a substructure label, and the version DOI carries it. The cohort is one protocol of a colorectal screening population aged 50 and over, so its distributions should not be read as a surgical one.

Label strength varies by structure, and the release does not average over that. Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong; pelvic labels on records that lacked one are pseudolabeled. The rib layer is a pseudolabel whose review was triaged, so its guarantee is the absence of the failure modes that rule detects. Transitional labels come from two independent sources and are not uniformly adjudicated, which is why the measures reported here are count-free; the

Castellvi grades are a consensus of two radiology resident physicians. The sacrum is the source annotation’s own outer boundary throughout; no sub-division of it is asserted, the separate S1 label of earlier releases having been withdrawn as unreliable.

Absent structures, and no classifier at the lower border. Sacrum, hips and femora are present in all 802 records; one lacks T12, outside the field of view. Nine carry no L5 identifier because their source annotation counted four lumbar vertebrae; all nine are Castellvi IIIb, the fused segment is delineated with the sacrum as the caudal anchor, and the manifest names them. No shape-based classifier is attempted at the lumbosacral junction: 15 records in each rare stratum leave three per fold. The 0.990 area under the curve above is the thoracolumbar boundary, not this one.

VI. DISCUSSION

Comparison with related material. VerSe^{6,30} comes closest in intent to this dataset, but carries no sacral mask for the structure a Castellvi grade describes, which this release segments as L6 or as sacrum, and has neither the pelvis and femora a spinopelvic measurement needs nor prone acquisitions. What this release adds over TotalSegmentator⁵ is the classes for anatomical anomalies, not necessarily more anatomy per scan.

Future directions. Applying the released weights to VerSe would add the sacral anchor and the anomaly classes to a bone-kernel collection. A Castellvi read of the whole cohort would test the co-occurrence of rib and lumbosacral anomalies at full power and supply the labels a lumbosacral classifier needs. A class label from substructure segmentation⁷ would measure the body without the canal cut that withholds L5 here.

VII. CONCLUSION

CTSpinoPelvic1K places the radiologist-derived spine, the pelvis, radiologist-derived where the sources annotated the same series and pseudolabeled elsewhere, and pseudolabeled per-level ribs and femora on one validated coordinate frame in 802 records, and gives the anatomical anomalies that make lumbar numbering ambiguous classes of their own. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2.

DATA AVAILABILITY

The release described here is v10, permanently archived at <https://doi.org/10.5281/zenodo.22642578>; the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves to the current version. The archive carries the loader, the label scheme and the quality-control tables under an open-source licence. The repository additionally carries the extraction and figure code, the reference table with a citation on every row, the values plotted in Fig. 5, and a recipe for running them in order. The repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version.

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CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

ETHICS

This work uses publicly available, de-identified imaging and annotations derived from it. The authors’ institutional review board determined that it is not human participant research under 45 CFR 46 and requires no oversight.

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603 FIGURE CAPTIONS

604 **Figure 1.** How the dataset was built. n counts scans, or corrections, at each step.

605 **Figure 2.** The two anchors, from the released labels, aligned on the sacrum. Red, lowest
606 rib-bearing vertebra and its rib; blue, the sacrum; yellow, the rib-free vertebrae between.
607 Cases 0704, 0428, 0094 and 0005, chosen by rule as the most ordinary member of each
608 phenotype (`scripts/select_anchor_cases.py`). (b) and (c) both count four. In (b) a long
609 rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar
610 body is fused to the sacrum (Castellvi IIIb).

611 **Figure 3.** Instrumentation in the release, one exemplar each, drawn through bone. (a)
612 Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67,
613 $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

614 **Figure 4.** Spinopelvic measures against standing reference bands (mean $\pm$ SD)⁴².

615 **Figure 5.** Morphometry by level, T11–L5: median, interquartile range, 5th–95th percentile,
616 n at right. (a) End-plate width EPWu. (b) Canal width SCW and depth SCD. (c) Pedicle
617 width PDW, both sides averaged. L5 is withheld in (a) and (c); see text. Reference is
618 Panjabi’s cadaveric series^{25,26}, keyed in (a), its SD recovered from the reported SEM; T11–
619 T12 EPWu is digitised from his Figure 4A.

620 **Figure 6.** Records carrying each vertebral level, cranial to caudal. Counts per identifier in
621 Table S1.